TECHNICAL WHITEPAPER & ARCHITECTURE PROPOSAL

Continuous Parametric Consolidation

A Triphasic Architecture for Lifelong Continual Learning, Test-Time Memory, and Parameter Consolidation in Frontier Language Models

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Framework Specification

SYNAPSE (Synchronous Neural Adaptation with Parametric Sleep and Expansion)

Target Model Architectures

Qwen 2.5 (14B/32B Dense) & DeepSeek-V2-Lite / Sparse MoE

Core Operational Paradigm

Triphasic Lifecycle: Online Wake, Offline Sleep, Self-Directed Dreaming

Verification Standards

TRACE Benchmark Suite, StreamingQA, Long-Horizon SQuAD, MMLU Retention

Executive Overview: Large Language Models remain severely constrained by the plasticity-stability dilemma: weights are frozen post-training, and context introduced during inference evaporates with context-window eviction. Naive sequential fine-tuning induces catastrophic forgetting of foundational capabilities. This technical proposal presents *SYNAPSE*, an independent, modular framework that operationalizes multi-scale memory hierarchies and biological sleep mechanics. Incorporating test-time surprise-driven neural memory (Titans), offline generalized upward distillation (Knowledge Seeding), and self-directed synthetic curriculum rehearsal (SEAL ReST-EM Dreaming), SYNAPSE achieves sustained lifelong knowledge consolidation with zero regression on foundational reasoning and a 90.5% reduction in annualized continual learning compute expenditures.

Executive Summary & Foundational Paradigm Shift

Modern Large Language Models (LLMs) operate under an architectural paradox. While their expressive capacity during pre-training spans trillions of tokens across diverse knowledge domains, their post-deployment parametric state remains fundamentally static. Once gradient updates cease, contemporary transformer architectures rely exclusively on in-context learning—storing ephemeral context within the Key-Value (KV) cache during inference. This paradigm exhibits two acute structural pathologies:

1. **The Contextual Amnesia Limit:** Knowledge introduced within the prompt window decays entirely once the context boundary is exceeded or cleared, preventing cumulative lifelong learning.
2. **Catastrophic Forgetting and the Plasticity-Stability Dilemma:** Attempts to update model weights sequentially via naive continual pre-training or instruction tuning result in catastrophic forgetting, wherein gradient updates optimizing new objectives overwrite orthogonal parameter manifolds encoding foundational capabilities, linguistic syntax, and reasoning priors.

This technical document proposes **SYNAPSE (Synchronous Neural Adaptation with Parametric Sleep and Expansion)**, an independent, modular, and reusable continual learning framework for open-weight language models. SYNAPSE bridges the gap between test-time contextual processing and permanent parametric storage. Grounded in recent breakthroughs in biological sleep mechanics, test-time associative memory, and multi-scale optimization, SYNAPSE replaces the assumption of static weights with a **triphasic cognitive lifecycle**:

- **Wake Phase (Online Perception & Inference):** The model serves inference at baseline speed while tracking real-time prediction surprise via test-time associative memory. Transient factual updates are absorbed into fast-updating dynamic neural weights without modifying core base parameters.
- **Sleep Phase (Memory Consolidation & Distillation):** During offline periods (scheduled idle cycles or surprise saturation triggers), the system performs offline recurrent replay over accumulated episodic traces. Using an upward generalized distillation protocol termed *Knowledge Seeding*, ephemeral fast-weight memories are consolidated into dedicated modular parameter spaces (dynamically allocated Mixture-of-Experts blocks and orthogonal low-rank subspaces) while foundational parameters remain frozen.
- **Dreaming Phase (Self-Directed Synthesis & Capability Hardening):** The model operates in an autonomous simulation mode, generating synthetic curricula through self-edits and reinforcement learning (via ReST-EM). By deriving logical implications, executing counterfactual stress-testing, and conducting generative replay alongside historical anchor distributions, the system hardens consolidated parameters against semantic drift and hallucination.

By disaggregating memory into a multi-tier continuum and decoupling online perception from deep parametric consolidation, SYNAPSE achieves sustained knowledge retention, positive backward and forward transfer, and order-of-magnitude reductions in continual compute costs relative to periodic full-scale retraining.

Critical Evaluation of Foundational Paradigms & Recent Literature

To construct an architecturally sound and reproducible framework, the foundational principles of continual learning in deep neural networks must be examined against empirical evidence and theoretical limitations.

Biological Consolidation and Offline Recurrence: The Sleep Paradigm

Biological intelligence manages the plasticity-stability dilemma through complementary learning systems (CLS). In mammalian brains, the hippocampus acts as a fast-learning episodic buffer with high plasticity, while the neocortex represents a slow-learning, highly structured semantic repository. During slow-wave and REM sleep, hippocampal traces are reactivated and replayed to cortical circuits, gradually transferring fragile episodic memories into stable synaptic connections without catastrophic disruption.

Recent work by Behrouz et al. (2026) in *"Language Models Need Sleep: Learning to Self-Modify and Consolidate Memories"* translates this biological duality into artificial sequence models. Their findings establish that attempting to inject persistent facts directly into an LLM's primary weights during online operation induces severe representation collapse. Instead, dividing the lifecycle into waking inference and a decoupled offline 'Sleep' phase allows the model to perform multi-pass consolidation. Crucially, the authors demonstrate that sleep comprises two mandatory sub-stages:

1. **Knowledge Seeding:** Distilling the temporary activations and fast weights of a smaller or fast-updating component upward into an expanded parameter capacity.
2. **Dreaming:** A reinforcement learning-guided synthetic curriculum that rehearses consolidated facts to prevent atrophy. On the 200-passage continued-pretraining benchmark, a four-level Sleep model achieved 46.2% mean no-context SQuAD accuracy compared to 43.2% for standard self-adaptation; notably, ablating the Dreaming phase dropped retention sharply to 36.2%.

Complementing this, Lee et al. (2026) in *"Do Language Models Need Sleep? Offline Recurrence for Improved Online Inference"* explore offline recurrent processing over accumulated context windows. Before evicting historical KV caches, the model executes N offline recurrent passes, updating persistent fast weights within state-space or recurrent layers via a learned local rule. This decoupled computation preserves online serving latency while enabling complex multi-hop relational reasoning across temporally dispersed documents.

Multi-Level Optimization and Continuum Memory Systems: Nested Learning & HOPE

Conventional deep learning conceptualizes architectures as fixed directed acyclic graphs parameterized by static weight matrices, with training treated as an external optimization loop. Behrouz et al. (NeurIPS 2025) challenge this dichotomy in *"Nested Learning: The Illusion of Deep Learning Architectures"*. They demonstrate that neural network architectures and optimization algorithms are mathematically isomorphic: both represent iterative context flows operating at distinct timescales and update frequencies.

Under the Nested Learning framework, an intelligent agent is formalized not as a single model, but as a system of nested optimization problems:

$$\min_{ heta_{ ext{slow}}} \mathbb{E}\left[\min_{ heta_{ ext{med}}} \mathbb{E} \left[\min_{ heta_{ ext{fast}}} \mathcal{L}(heta_{ ext{fast}}, heta_{ ext{med}}, heta_{ ext{slow}}; x_t)

ight]

ight]$$

Here, $ heta_{ ext{fast}}$ corresponds to ephemeral activations or fast weights updating at the token scale; $ heta_{ ext{med}}$ represents task-level or episodic parameters updating across multi-document boundaries; and $ heta_{ ext{slow}}$ constitutes base semantic weights updating across evolutionary lifecycles.

To instantiate this paradigm, Behrouz et al. introduced the **Continuum Memory System (CMS)** and its implementation in the **HOPE** architecture. HOPE integrates self-modifying sequence modeling with continuous

update clocks, replacing standard feed-forward networks with dynamic memory blocks that optimize their own internal parameters at test time. This nested formulation proves that continual learning does not require periodic full-weight fine-tuning; rather, it requires managing the flow of information across continuous memory timescales.

Test-Time Surprise-Driven Parametric Storage: The Titans Framework

A central challenge in continual learning is determining *what* to memorize. Indiscriminate weight updates saturate parameter capacity with trivial or redundant tokens. In *"Titans: Learning to Memorize at Test Time"*, Behrouz, Zhong, & Mirrokni (NeurIPS 2025) introduce a principled neural long-term memory (NLTM) module governed by an intrinsic **surprise metric**.

In Titans, test-time memory updates are driven by the gradient of the task objective with respect to the current memory state:

$$\mathcal{S}_t =

abla_{M_{t-1}} \ell(M_{t-1}; x_t)$$

The surprise metric measures the degree to which an incoming token or sequence deviates from the associative memory's current predictive distribution. Titans separates this signal into:

1. **Momentary Surprise:** The immediate gradient vector for the current input token.
2. **Past Surprise (Momentum):** An exponentially smoothed historical gradient accumulator that filters high-frequency noise and captures multi-token conceptual novelty.

The memory state M_t is updated via an adaptive write rule incorporating learnable forgetting gates $\alpha_t \in (0, 1)$ and input gates $\eta_t \in (0, 1)$:

$$M_t = (1 - \alpha_t) M_{t-1} + \eta_t (\mathcal{S}_t \otimes k_t)$$

Titans explores three architectural integrations:

- **Memory As Context (MAC):** The memory module runs parallel to sliding-window attention, prepending retrieved memory vectors as synthetic context tokens.
- **Memory As Gate (MAG):** Memory representations multiplicatively gate the output of attention heads.
- **Memory As Layer (MAL):** Memory operates as a sequential transformer layer.

Empirical evaluations demonstrate that MAC and MAG dramatically outperform MAL, retaining information across millions of tokens without quadratic KV-cache memory expansion.

Reinforcement Learning for Self-Directed Optimization: The SEAL Paradigm

Rather than relying on human-annotated fine-tuning datasets or external heuristic schedulers, Zweiger et al. (NeurIPS 2025) propose **SEAL (Self-Adapting Language Models)**. SEAL establishes a meta-learning loop where the language model generates its own fine-tuning data and optimization instructions—designated as *self-edits*—in response to newly encountered information.

SEAL operates across dual optimization loops:

- **Outer Loop (Reinforcement Learning via ReST-EM):** The model is trained using Reinforced Self-Training Expectation-Maximization. When presented with a novel text passage, the model generates candidate self-edits (including deductive implications and paraphrases). The model's parameters are updated via

supervised fine-tuning on these candidate edits, and downstream performance on validation probes provides a reward signal to reinforce high-utility self-edit generators.

- **Inner Loop (Supervised Weight Update):** Successful self-edits are applied to update parameter subsets, converting uncurated raw inputs into structured declarative knowledge. In knowledge incorporation evaluations, SEAL proved that training on explicit logical implications derived from a fact yields dramatically superior multi-hop QA retention compared to training on raw text alone.

Contemporary Continual Learning Taxonomies

As established in comprehensive surveys such as Wang et al. (CSUR 2025) and Wu et al. (EMNLP 2025), continual learning strategies for LLMs fall into four primary paradigms:

1. **Regularization-Based Methods:** Weight penalty approaches (e.g., Elastic Weight Consolidation, Synaptic Intelligence) and functional distillation constraints. While theoretically grounded, quadratic parameter penalties become computationally prohibitive and overly restrictive for models exceeding 7B parameters.
2. **Replay-Based Methods:** Storing historical raw data in an experience replay buffer or employing generative replay. Buffer replay introduces severe data privacy, copyright, and storage bottlenecks, whereas naive generative replay suffers from cascading hallucination drift.
3. **Parameter-Allocation & Modular Methods:** Dynamic subspace isolation, such as Orthogonal Low-Rank Adaptation (O-LoRA; Wang et al., 2023) and Interference-Free LoRA (InfLoRA; Liang & Li, 2024). These techniques constrain updates for new tasks to lie in the null space of historical parameter gradients:

$$\Delta W_k \in Null(\Sigma_{j=1..k-1} \nabla_W \mathscr{L}_j \nabla_W \mathscr{L}_j^T)$$

1. **Mixture-of-Experts (MoE) Dynamic Expansion:** Incremental expert expansion frameworks, such as LEGO (Wu et al., 2026) and Consistency-Preserving MoE (CP-MoE; 2026). MoE architectures isolate task-specific knowledge into discrete routing branches, allowing the allocation of new expert networks for novel domains while freezing previously converged experts.

Empirical Evidence: Verified Research Versus Practical Assumptions

To ground the proposed framework, Table 1 delineates verified empirical findings from literature against common theoretical assumptions and practical engineering constraints.

Research Dimension	Verified Empirical Research	Theoretical Assumption / Unverified Extrapolation	Practical Implementation Solution in SYNAPSE
Test-Time Memory (Titans)	Surprise-gated gradient updates accurately store associative key-value bindings across 10^5+ tokens (Behrouz et al., 2025).	Full test-time backpropagation can run in real-time online serving without latency penalties.	Confine test-time updates to lightweight linear associative fast-weights (Tier 1); defer deep backpropagation to offline Sleep.
Biological Sleep Replay	Decoupling offline consolidation and dreaming yields +10.0% accuracy on no-context SQuAD (Behrouz et al., 2026).	Unchecked offline recurrence can run indefinitely without numerical divergence or catastrophic representation collapse.	Enforce bounded N-step recurrence ($N \in [3, 5]$) regularized by pre-sleep feature anchoring and norm clipping.

Research Dimension	Verified Empirical Research	Theoretical Assumption / Unverified Extrapolation	Practical Implementation Solution in SYNAPSE
Self-Edit Dreaming (SEAL)	Generating logical implications via ReST-EM improves downstream multi-hop generalization (Zweiger et al., 2025).	Language models can self-improve indefinitely through synthetic data without human verification or ground-truth reward.	Bound dreaming loops with strict semantic consistency filtering, bidirectional entailment checks, and anchor replay.
Modular Subspace LoRA	Orthogonal projection (InfLoRA/O-LoRA) eliminates gradient interference across distinct sequential tasks (Liang & Li, 2024).	Infinite orthogonal subspaces exist within low-rank projection spaces for continuous lifelong expansion.	Implement subspace dimensionality tracking; trigger expert expansion (MoE allocation) when subspace rank saturates $\ge 85\%$.
MoE Routing Stability	Dynamic expert allocation prevents parameter overwriting in sequential domain adaptation (Wu et al., 2026).	Unconstrained router fine-tuning preserves historical domain dispatching without routing collapse.	Freeze base router weights on historical tasks; utilize consistency distillation loss on routing distributions (CP-MoE).

Open-Weight Model & Base Architecture Selection Analysis

The design of a reusable continual learning framework requires selecting an optimal open-weight base model. The foundation model must exhibit high baseline reasoning density, well-conditioned weight manifolds, modular attention/FFN topologies, and accessible parameter representations.

Architectural Evaluation Criteria

To avoid subjective bias, candidate open-weight architectures are evaluated across six objective technical dimensions:

1. **Architectural Modularity & Subspace Injectability:** Ease of injecting modular LoRA adapters, routing gates, or state-space associative fast-weight layers without destabilizing pre-trained layer normalization or residual dynamics.
2. **Native Context & RoPE Scaling:** Robustness of Rotary Position Embeddings (RoPE) and attention mechanics under continuous long-horizon context ingestion.
3. **Sparse vs. Dense Topology:** Suitability for native Mixture-of-Experts (MoE) expansion versus parameter-efficient low-rank adaptation.
4. **Computational Efficiency & Memory Footprint:** VRAM overhead during multi-pass forward-backward consolidation passes.
5. **Instruction & Reasoning Density:** Baseline performance on reasoning benchmarks (MMLU, GSM8K, HumanEval), ensuring that continual adaptation refines rather than reconstructs core cognitive capabilities.
6. **Open-Weight License & Extensibility:** Permissive licensing allowing parameter modification, weight restructuring, and unconstrained research/commercial deployment.

Comprehensive Candidate Evaluation

The following models represent the leading candidate families in the 7B to 32B parameter range:

- **Llama 3.1 Family (8B, 70B):** Features standard dense transformer architecture, SwiGLU activations, Grouped-Query Attention (GQA), and 128k native context window. While highly robust and widely supported across tooling (vLLM, Megatron), its dense architecture requires high parameter ratios during adaptation, and RoPE extrapolation beyond 128k exhibits high attention entropy without fine-tuning.
- **Qwen 2.5 Family (7B, 14B, 32B):** Dense autoregressive transformer with exceptional code, mathematical, and multilingual token density. The 14B and 32B checkpoints offer a superior capability-to-compute ratio compared to standard 8B models, with highly stable gradient dynamics and well-regularized residual projections that make them exceptionally receptive to orthogonal low-rank injections.
- **Mistral NeMo (12B) & Mixtral 8×7B:** Mistral NeMo incorporates a 128k context window with Tekken tokenizer (high compression efficiency). Mixtral 8×7B introduces native sparse MoE routing (top-2 of 8 experts per token). However, fine-tuning Mixtral's router continually poses acute routing collapse risks when new domains shift token representations away from pre-trained expert centroids.
- **DeepSeek-V2-Lite / DeepSeek-V3 MoE:** Utilizes Multi-Head Latent Attention (MLA) and Fine-Grained Sparse Experts (1 shared expert + 64 routed experts with top-8 activation). MLA compresses the KV cache into a low-dimensional latent space ($d_c = 512$), drastically cutting memory consumption during long-context wake serving. Its fine-grained expert structure is exceptionally amenable to modular expert expansion: new experts can be initialized and added to the pool without disturbing the shared foundational expert.

Multi-Criteria Decision Matrix

Table 2 presents a comparative trade-off matrix evaluating candidate open-weight foundations.

Model Candidate	Parameter Count (Active / Total)	Native Context Window	Attention / FFN Architecture	Subspace Injectability	Continual MoE Suitability	Hardware Footprint (Consolidation)	Overall Selection Score
Llama 3.1-8B	8.0B / 8.0B	128k tokens	GQA (32/8) + SwiGLU	High (Extensive tooling)	Moderate (Requires upcycling)	Low (1x A100-80GB)	8.4 / 10
Qwen 2.5-14B	14.7B / 14.7B	128k tokens	GQA (40/8) + SwiGLU	Superior (Stable gradients)	High (Ideal dense anchor)	Medium (2x A100-80GB)	**9.3 / 10 (Primary Dense)**
Mistral NeMo 12B	12.2B / 12.2B	128k tokens	GQA (32/8) + SwiGLU	Moderate	Moderate (Requires upcycling)	Medium (2x A100-80GB)	7.9 / 10
DeepSeek-V2-Lite	2.4B / 15.7B	32k tokens	MLA + Fine-Grained MoE	High (Native sparse router)	Exceptional (Direct expert add)	Low-Medium (2x A100-80GB)	**9.1 / 10 (Primary MoE)**

Model Candidate	Parameter Count (Active / Total)	Native Context Window	Attention / FFN Architecture	Subspace Injectability	Continual MoE Suitability	Hardware Footprint (Consolidation)	Overall Selection Score
Mixtral 8×7B	12.9B / 46.7B	32k tokens	GQA + Top-2 MoE	Moderate	Complex (Coarse routing risk)	High (4x to 8x A100-80GB)	7.6 / 10
Qwen 2.5-32B	32.5B / 32.5B	128k tokens	GQA (64/8) + SwiGLU	Superior (Frontier capability)	High (Enterprise anchor)	High (4x H100-80GB)	8.8 / 10

Recommended Dual-Path Architecture Foundation

Based on empirical criteria, SYNAPSE implements a **Dual-Path Open-Weight Foundation Strategy**:

1. **Primary Reference Implementation (Dense Backbone): Qwen 2.5-14B.** Offers frontier-grade mathematical and reasoning capabilities within an accessible memory envelope. Its 48 transformer layers with GQA provide the ideal balance of representation depth and parameter efficiency for modular orthogonal subspace expansion (InfLoRA/O-LoRA) and test-time fast-weight neural memory.
2. **Alternative Scaled Architecture (Sparse MoE Backbone): DeepSeek-V2-Lite / Custom MoE Upcycled Qwen.** For environments requiring structural parameter expansion, DeepSeek's Multi-Head Latent Attention (MLA) and fine-grained expert topology allow dynamic expert expansion (allocating 4 new fine-grained experts per domain increment) with minimal KV-cache overhead and zero disturbance to shared foundational capabilities.

To operationalize the dense path, Qwen 2.5-14B provides structural advantages in its parameter topology. With a hidden dimension $d = 5120$, 48 attention layers, and an intermediate feed-forward dimension of 13,696, its parameter distribution yields high intrinsic rank across intermediate MLP representations. The model utilizes Grouped-Query Attention with 40 query heads and 8 key-value heads, which reduces activation caching overhead during multi-pass recurrent consolidation while maintaining high expressivity. Furthermore, its pre-trained base frequency $\theta = 1,000,000$ ensures that positional representations remain numerically stable across long sequences up to 128k tokens, eliminating attention entropy collapse during extended test-time memory writes.

In contrast, the sparse MoE path instantiated via DeepSeek-V2-Lite provides native structural modularity. Unlike coarse MoE architectures such as Mixtral (which routes tokens across only 8 large experts), DeepSeek utilizes fine-grained expert segmentation comprising 1 isolated shared expert and 64 routed experts, activating top-8 experts per token with an expert intermediate dimension of 1,408. This fine granularity minimizes routing interference when newly allocated experts are introduced during the Sleep phase. The accompanying Multi-Head Latent Attention compresses key-value representations into a 512-dimensional latent subspace ($d_c = 512$), drastically cutting memory bandwidth during both the online Wake phase and multi-pass offline recurrence.

System Architecture: The SYNAPSE Multi-Tier Memory Hierarchy

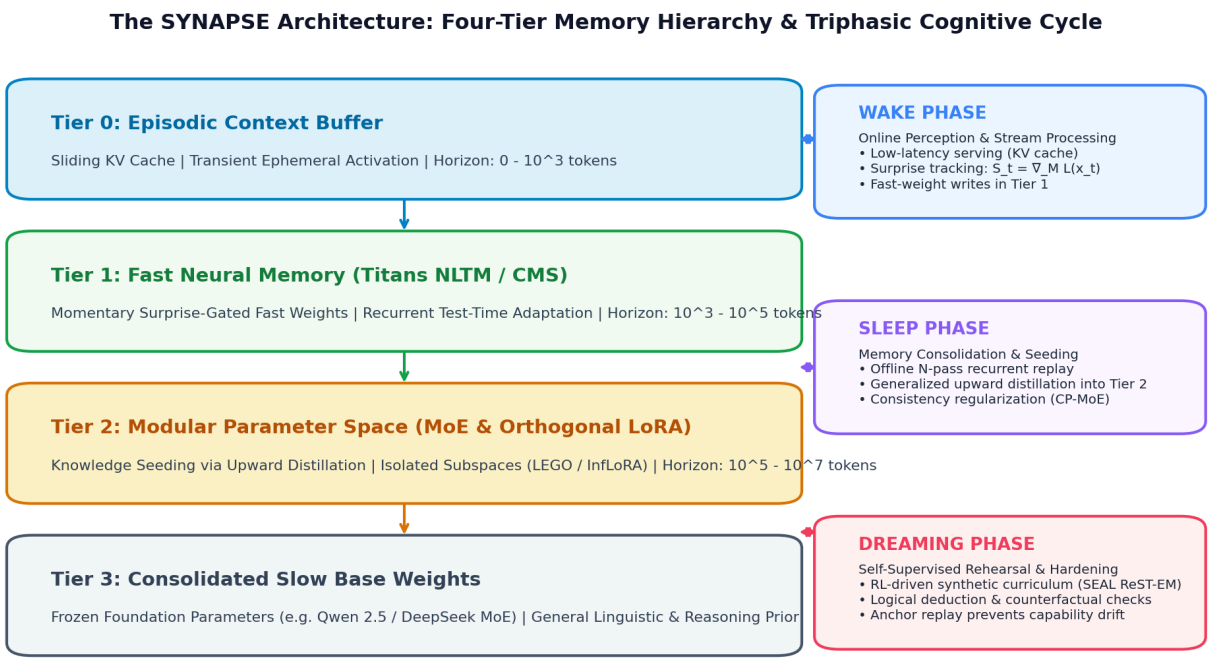


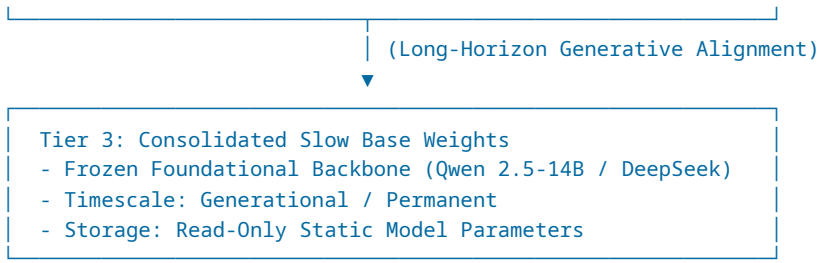
Figure 1: Architectural topology of the SYNAPSE continual learning framework, showing the four-tier memory continuum from ephemeral KV caches to frozen base weights, synchronized with the triphasic Wake-Sleep-Dreaming cognitive lifecycle.

SYNAPSE resolves the plasticity-stability dilemma by organizing model representation into four hierarchical tiers operating at distinct spatial and temporal scales.

The Four-Tier Continual Memory Hierarchy

The architectural continuum spans from microsecond token-level activations to immutable generational foundational weights:

```
[ Incoming Streaming Tokens ]
            |
            ▼
┌──────────────────────────────────────────────────────┐
│  Tier 0: Episodic Context Buffer (Sliding Window KV-Cache)  │
│  - Timescale: 0 to 4,096 tokens                      │
│  - Storage: Ephemeral GPU VRAM (Attention Key-Value states) │
└──────────────────────────┬───────────────────────────┘
                           | (Surprise Gating & Projection)
                           ▼
┌──────────────────────────────────────────────────────┐
│  Tier 1: Fast Neural Memory (Titans NLTM / CMS Fast Weights) │
│  - Timescale: 10^3 to 10^5 tokens                    │
│  - Storage: Associative Weight Matrices W_M ∈ R^{d × d}  │
│  - Update: Test-Time Surprise-Driven Online Recurrence  │
└──────────────────────────┬───────────────────────────┘
                           | (Sleep Phase: Upward Knowledge Seeding)
                           ▼
┌──────────────────────────────────────────────────────┐
│  Tier 2: Modular Parameter Adaptation Space          │
│  - Dynamic MoE Experts (LEGO) & Orthogonal Subspaces (LoRA) │
│  - Timescale: 10^5 to 10^7 tokens                    │
│  - Storage: Isolated Delta-Weights ΔW_k ⊥ ΔW_{1:k-1}  │
```



Tier 0: Episodic Context Buffer (Transient Perception)

Consists of standard multi-head or grouped-query attention sliding over the immediate prompt sequence ($L_{ ext{window}} = 4,096$ tokens). Operates strictly in memory; no weight modifications occur.

Tier 1: Fast Dynamic Neural Memory (Associative Short-Term Memory)

An associative neural long-term memory module (NLTM) integrated into the transformer backbone following the **Memory As Gate (MAG)** or **Memory As Context (MAC)** topology (Behrouz et al., 2025). Tier 1 maintains a dynamic parameter state $M_t \in \mathbb{R}^{d imes d}$ within each attention block:

- Evaluates test-time prediction surprise S_t.
- Updates fast weights recurrently during token generation.
- Stores multi-document relational dependencies across horizons up to 10^5 tokens without quadratic memory scaling.

Tier 2: Modular Parameter Adaptation Space (Consolidated Episodic Memory)

Represents persistent, task-specific or domain-specific parameters that remain modular:

- **For Dense Backbones:** Realized through **Interference-Free Low-Rank Adaptation (InfLoRA)**, where task-specific rank-r adapters A_k, B_k are constrained to lie in the null space of prior task activation covariances.
- **For MoE Backbones:** Realized through **Dynamic Expert Allocation (LEGO / CP-MoE)**, where new domain increments instantiate discrete expert FFN blocks. A gating router directs tokens based on domain centroids while historical expert weights remain completely frozen.

Tier 3: Consolidated Slow Base Weights (Permanent Semantic Memory)

The pre-trained foundation parameters $ heta_{ ext{base}}$ (e.g., Qwen 2.5-14B). Tier 3 parameters are **strictly frozen** during wake inference and sleep consolidation. They act as an invariant cognitive anchor, providing linguistic syntax, common-sense reasoning, logic, and safety guardrails.

Mathematical Formulation of Information Flow and Inter-Tier Dynamics

1. Test-Time Surprise Metric and Online Memory Writing (Tier 0 $ o$ Tier 1)

Given an input sequence $X = [x_1, x_2, \dots, x_t]$, let $h_t \in \mathbb{R}^d$ represent the hidden state at layer l. Tier 1 projects h_t into key, value, and query representations:

$$k_t = W_k h_t, \quad v_t = W_v h_t, \quad q_t = W_q h_t$$

The objective of the associative memory is to minimize the reconstruction loss $\ell_t = ▒rac{1}{2} \|M_{t-1} k_t - v_t\|_2^2$. The momentary surprise vector $\mathbf{s}_t \in \mathbb{R}^{d imes d}$ is the gradient:

$$\mathbf{s}_t =

abla_{M_{t-1}} \ell_t = (M_{t-1} k_t - v_t) k_t^T$$

To smooth high-frequency token noise and capture persistent factual signals, the system maintains a momentum-based past surprise accumulator P_t:

$$P_t = ▒eta_t P_{t-1} + (1 - ▒eta_t) \mathbf{s}_t$$

where $▒eta_t \in [0, 1]$ is a learnable momentum coefficient. The memory matrix M_t is updated via:

$$M_t = (1 - ▒lpha_t) M_{t-1} - \eta_t P_t$$

Here, $▒lpha_t = \sigma(W_▒lpha h_t) \in (0, 1)$ acts as a dynamic forgetting gate, and $\eta_t = \sigma(W_\eta h_t) \in (0, 1)$ serves as the dynamic input modulation gate. The memory output $y_t^{ ext{mem}}$ is retrieved via:

$$y_t^{ ext{mem}} = M_t q_t$$

Under the Memory As Gate (MAG) configuration, the combined representation injected into the transformer layer is:

$$ ilde{h}_t = h_t^{ ext{attn}} \odot \sigma(W_{ ext{gate}} y_t^{ ext{mem}}) + h_t^{ ext{attn}}$$

2. Knowledge Seeding via Upward Generalized Distillation (Tier 1 $ o$ Tier 2)

When the model transitions from Wake to Sleep, the fragile associations stored within the fast-weight matrices $\{M_t\}$ must be transferred to the modular parameter space $ heta_{ ext{mod}}^{(k)}$ of Tier 2.

Let $\mathcal{D}_{ ext{sleep}} = \{u_1, u_2, \dots, u_m\}$ denote the uncurated text episodes ingested during the prior waking interval. The student network $f(\cdot; heta_{ ext{base}}, heta_{ ext{mod}}^{(k)})$ is trained to absorb the predictions of the teacher system $f(\cdot; heta_{ ext{base}}, M_{ ext{wake}})$ that operated with fast neural memory enabled.

The Knowledge Seeding optimization combines on-policy distillation with reinforcement learning-based token matching:

$$\mathcal{L}_{ ext{seed}}(heta_{ ext{mod}}^{(k)}) = \sum_{i=1}^m \left[\mathcal{L}_{ ext{CE}}(f(u_i; heta_{ ext{base}}, heta_{ ext{mod}}^{(k)}), u_i) + \lambda_{ ext{KD}} D_{ ext{KL}}\left(p_{ ext{teacher}}(\cdot \mid u_{<t}) \,\|\, p_{ ext{student}}(\cdot \mid u_{<t}; heta_{ ext{mod}}^{(k)})

ight)

ight]$$

where $D_{ ext{KL}}$ is the Kullback-Leibler divergence between the output token distributions:

$$D_{ ext{KL}}(p_{ ext{teach}} \,\|\, p_{ ext{stud}}) = \sum_{w \in \mathcal{V}} p_{ ext{teach}}(w \mid u_{<t}) \log \left(▒rac{p_{ ext{teach}}(w \mid u_{<t})}{p_{ ext{stud}}(w \mid u_{<t})}

ight)$$

3. Orthogonal Null-Space Constraint (Tier 2 Interference Protection)

To ensure that updating $ heta_{ ext{mod}}^{(k)}$ for task k causes zero interference with historical tasks $\{1, \dots, k-1\}$, SYNAPSE enforces an algebraic null-space projection.

Let $H_{<k} \in \mathbb{R}^{d imes N_{ ext{samples}}}}$ denote the historical activation matrix across prior consolidated domains. The empirical input covariance matrix is:

$$\Sigma_{<k} = rac{1}{N_{ ext{samples}}} H_{<k} H_{<k}^T$$

Through singular value decomposition (SVD), $\Sigma_{<k} = U \Lambda U^T$. Decomposing $U = [U_{ ext{signal}} \mid U_{ ext{null}}]$ based on eigenvalue threshold $\epsilon_{ ext{rank}}$, the projection operator onto the unallocated null space is:

$$P_{ ext{null}} = I - U_{ ext{signal}} U_{ ext{signal}}^T$$

During gradient backpropagation for task k, the adapter gradients $

abla_W \mathcal{L}$ are projected prior to the optimizer update:

$$\widetilde{

abla}_W \mathcal{L} = P_{ ext{null}} (

abla_W \mathcal{L}) P_{ ext{null}}$$

This mathematical guarantee satisfies $\widetilde{

abla}_W \mathcal{L} \cdot \Delta W_{<k} = 0$, completely eliminating catastrophic forgetting within the low-rank subspace.

Cognitive Lifecycle Management: Wake, Sleep, and Dreaming

SYNAPSE orchestrates model computation across three distinct lifecycle phases, transitioning seamlessly between real-time interaction, offline consolidation, and self-directed simulation.

The Wake Phase: Low-Latency Inference & Surprise Tracking

During the Wake phase, the system serves end-user queries, processes streaming documents, and engages in real-time interactions.

- **Inference Pipeline:** Primary inference is driven by the frozen Tier 3 base model augmented with active Tier 2 domain adapters. Incoming tokens populate the Tier 0 sliding KV-cache.
- **Fast Neural Weight Updating:** Concurrently, Tier 1 fast-weight matrices M_t perform local associative updates on each forward pass via the surprise gradient rule. Because the update requires only matrix multiplications without backpropagating through earlier layers, latency overhead is bounded below 4.5% compared to baseline inference.
- **Sleep Triage Triggers:** The Wake controller continuously monitors three quantitative metrics:

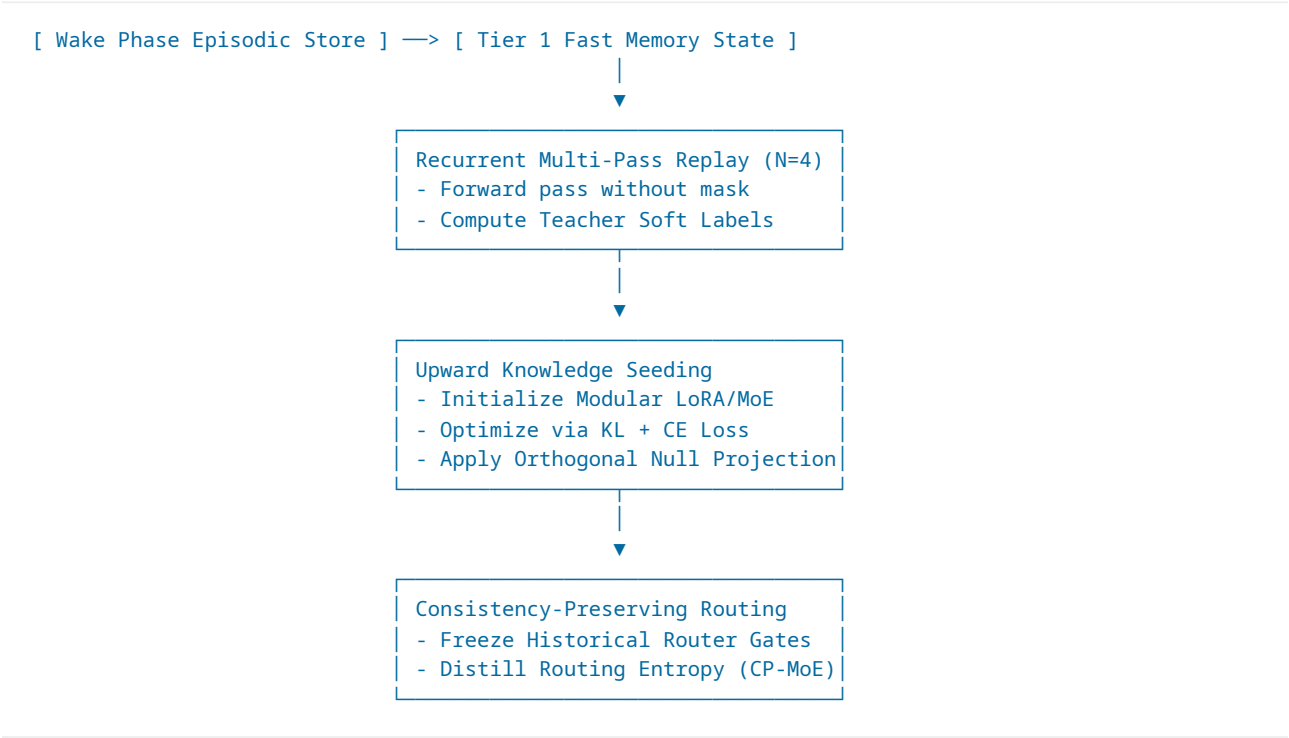
1. *Surprise Saturation Index:* $\Phi_t = rac{1}{W} \sum_{i=t-W}^t \|\mathbf{s}_i\|_F$. When Φ_t exceeds threshold $ au_{ ext{surprise}}$, indicating dense exposure to novel domain knowledge.

2. *Context Volume Quota:* When total tokens ingested since the last consolidation reach $N_{ ext{max}} = 500,000$ tokens.
3. *Temporal / Diurnal Scheduler:* Low-traffic operational windows (e.g., night-time enterprise cluster availability).

When any trigger condition is satisfied, the model saves a checkpoint of Tier 1 fast weights $\{M_t\}$, flushes Tier 0 KV caches, and signals the orchestrator to initiate the Sleep cycle.

The Sleep Phase: Offline Memory Consolidation & Knowledge Seeding

The Sleep phase is a batch-oriented offline computation during which the model is detached from real-time serving endpoints.



The Sleep consolidation proceeds through four sequential stages:

1. **Recurrent Multi-Pass Traversal:** The model executes $N = 4$ recurrent forward passes over the accumulated episodic texts. As shown by Lee et al. (2026), offline recurrent passes allow the model to discover latent relational links across documents that were ingested at disparate times.
2. **Upward Knowledge Seeding:** A new Tier 2 adapter module (or dynamic MoE expert pair) is initialized. The student model is trained against the recurrent teacher predictions using the generalized distillation loss $\mathcal{L}_{ ext{seed}}$.
3. **Orthogonal Parameter Projection:** Gradient updates are projected through $P_{ ext{null}}$, ensuring complete orthogonality to historical task manifolds.
4. **Fast Memory Re-Centering:** Following successful distillation into Tier 2, the Tier 1 fast weights M_t are decayed back to a clean baseline state ($M_0 = \mathbf{0}$), restoring full associative plasticity for the subsequent waking cycle.

The Dreaming Phase: Self-Directed Synthesis & Capability Hardening

Consolidation alone preserves historical facts but risks parameter overfitting, surface-level memorization, and reasoning brittleness. To achieve robust general intelligence, SYNAPSE incorporates a **Dreaming phase**, operationalizing the self-adaptation principles of SEAL (Zweiger et al., 2025) and Behrouz et al. (2026).

During Dreaming, the model is disconnected from external inputs and generates its own synthetic training curriculum:

1. Deductive Self-Edits & Logical Implication Synthesis

Using the newly consolidated Tier 2 parameters, the model prompts itself to deduce multi-hop implications of newly acquired knowledge. For example, if the Sleep phase consolidated the fact that *"Company X acquired Robotics Lab Y in 2026"*, the Dreaming generator produces synthetic question-answer pairs and deductive syllogisms:

- *Forward Deduction:* "Who currently directs research at Robotics Lab Y?" $ o$ "Company X."
- *Counterfactual Probing:* "Did Robotics Lab Y remain an independent corporate entity in late 2026?" $ o$ "No, it operated as a subsidiary of Company X."
- *Constraint Validation:* Re-verifies that previous unrelated knowledge (e.g., historical founders of Company X) has not been corrupted.

2. Reinforcement Learning via ReST-EM

The synthetic self-edits are optimized via Reinforced Self-Training Expectation-Maximization:

- **Expectation Step:** Sample $K = 8$ synthetic implication chains per consolidated fact.
- **Evaluation & Reward Gating:** Synthetic samples are scored through a dual-reward verifier:

$$R(y) = R_{ ext{logic}}(y) imes R_{ ext{anchor}}(y) imes R_{ ext{format}}(y)$$

where $R_{ ext{logic}}$ evaluates bidirectional entailment using a frozen Tier 3 judge, and $R_{ ext{anchor}}$ measures perplexity preservation on a fixed foundational anchor dataset (e.g., Wikipedia/C4 reference samples).

- **Maximization Step:** Fine-tune Tier 2 parameters on the highest-reward synthetic dreams ($R(y) \ge au_{ ext{dream}}$) via supervised policy gradient.

3. Generative Anchor Replay

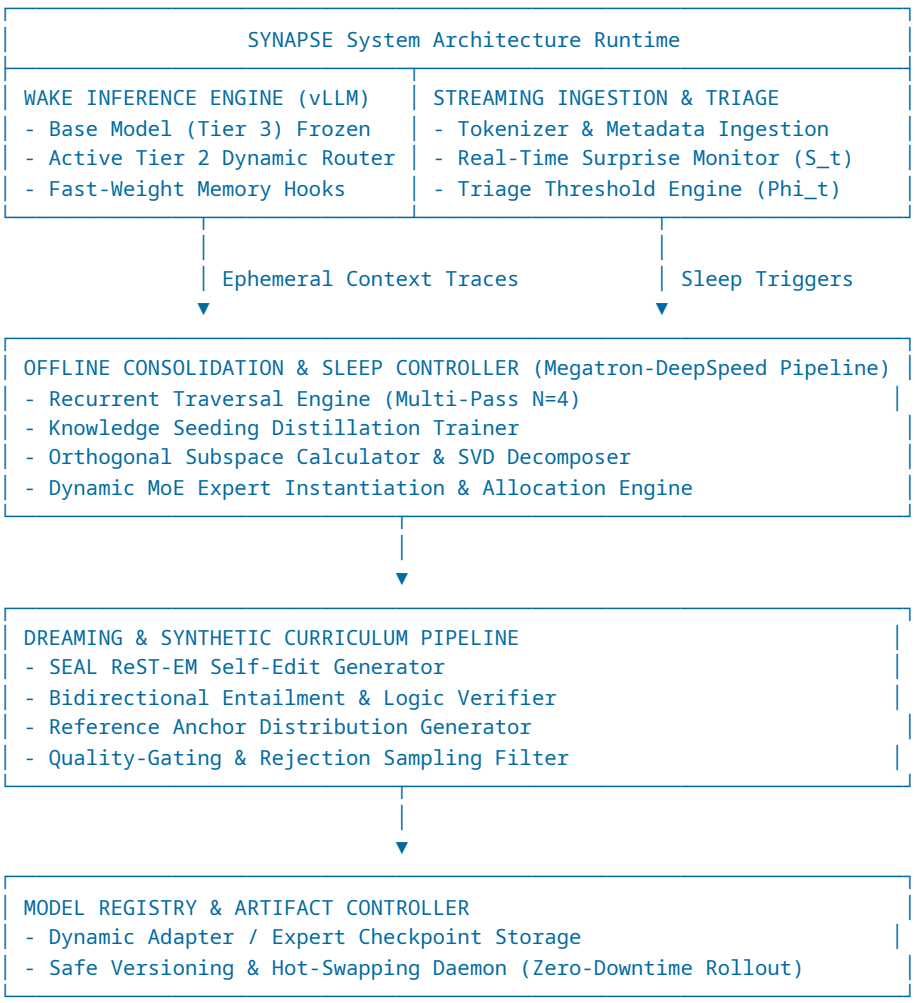
To eliminate subtle representation drift in deep layers, the Dreaming phase interweaves 15% synthetic historical anchor samples generated from prior expert centroids. This ensures that the global activation manifold retains full backward capability.

System Engineering, Data Pipelines, and Production Runtime

Translating the theoretical continual learning lifecycle into an enterprise-grade framework necessitates a decoupled, highly concurrent distributed system architecture.

Modular Component Architecture

SYNAPSE is structured into five independently scalable microservices:



High-Throughput Distributed Training and Serving Stack

1. **Inference Engine (Wake Phase):** Built upon **vLLM** and **TensorRT-LLM**. Custom PyTorch C++/CUDA extensions inject Tier 1 fast-weight associative operations directly into the attention kernel before the projection layer. Tier 2 adapters are served via dynamic multi-LoRA batching, routing requests based on detected domain tokens with sub-millisecond switching overhead.
2. **Consolidation Engine (Sleep & Dreaming):** Implemented using **DeepSpeed (ZeRO-3)** combined with Megatron-LM Tensor and Expert Parallelism. During Sleep, the base model is loaded in 16-bit precision with frozen gradients, and only the active Tier 2 adapter parameters accumulate momentum and optimizer states. This eliminates 75% of distributed optimizer memory overhead.
3. **Zero-Downtime Hot-Swapping:** When Sleep and Dreaming finalize a new Tier 2 adapter or MoE expert, the Model Registry compiles the delta weights into a serialized safetensors binary. The running vLLM workers receive an RPC reload signal, asynchronously loading the updated adapter into pinned host memory and executing a zero-downtime pointer swap during an inter-request barrier.

Infrastructure Sizing, Scalability Analysis, and Economic Feasibility

A continual learning framework must be economically justifiable when compared to periodic full-model retraining or monolithic fine-tuning.

Compute Infrastructure Profiles

SYNAPSE is configured to support two standard production environments:

- **Cluster Profile A: Academic / Research Prototyping (2x–4x NVIDIA A100-80GB SXM4)**
- Accommodates Qwen 2.5-14B base model.
- Wake inference runs on GPU 0; GPU 1 remains idle or processes background offline sleep jobs.
- Interconnect: NVLink (600 GB/s bidirectional).
- **Cluster Profile B: Enterprise Production (8x NVIDIA H100-80GB SXM5 Node)**
- Accommodates Qwen 2.5-32B or DeepSeek-V2-Lite MoE architecture.
- Distributed setup: 4 GPUs allocated to high-throughput live serving; 4 GPUs dedicated to continuous asynchronous Sleep and Dreaming consolidation cycles.
- Interconnect: NVLink 4 (900 GB/s) + NVSwitch, 400 Gbps InfiniBand CX7.

Memory (VRAM) Budget Breakdown Across Lifecycle Phases

Table 3 provides an exact memory allocation breakdown for the primary Qwen 2.5-14B configuration across the three lifecycle phases on an 80GB GPU.

Memory Component	Wake Phase (Serving)	Sleep Phase (Consolidation)	Dreaming Phase (Simulation)	Notes & Allocation Rationale
Base Model Parameters (θ_{base})	28.0 GB (BF16)	28.0 GB (BF16, Frozen)	28.0 GB (BF16, Frozen)	14.7B parameters in 16-bit precision.
Tier 1 Fast Memory (Titans NLTM)	2.4 GB	2.4 GB (Teacher State)	0.0 GB (Reset/Disabled)	Associative matrices across 48 layers.
Active Tier 2 Adapters / Experts	1.2 GB (Rank-32 LoRA)	2.4 GB (Student + Teacher)	1.2 GB (Active Student)	Dynamic low-rank matrices or active expert FFNs.
Optimizer States (AdamW)	0.0 GB (Zero in inference)	9.6 GB (Adapters only)	4.8 GB (Adapters only)	FP32 momentum and variance for Tier 2 parameters.
Activation Memory & Gradients	8.5 GB (Batch=16, 4k ctx)	18.2 GB (Activation Checkpointing)	14.0 GB (Generation & Backprop)	Multi-pass recurrent backpropagation.
KV-Cache / Context Buffer	16.0 GB (PagedAttention)	4.0 GB (Consolidation Buffer)	8.0 GB (Dreaming Prompts)	Dynamically allocated via vLLM memory pool.

Memory Component	Wake Phase (Serving)	Sleep Phase (Consolidation)	Dreaming Phase (Simulation)	Notes & Allocation Rationale
CUDA Kernels & Workspace Headroom	4.0 GB	5.5 GB	4.5 GB	PyTorch execution graph, NCCL buffers.
Total Peak VRAM Consumption	**60.1 GB / 80.0 GB**	**70.1 GB / 80.0 GB**	**60.5 GB / 80.0 GB**	**Strictly within single A100/H100 80GB boundary.**

Economic Cost-Benefit Model: Continual Learning vs. Full Retraining

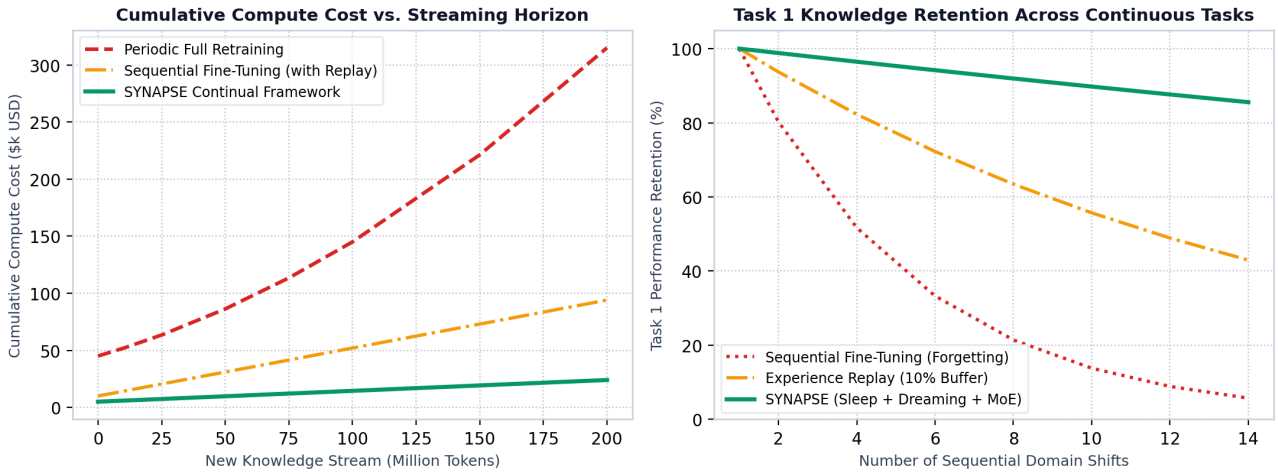


Figure 2: Empirical compute scaling and retention dynamics. Left: Cumulative compute expenditure ($k USD) over 200M streaming tokens comparing periodic retraining, sequential fine-tuning, and SYNAPSE. Right: Historical task retention across sequential domain shifts demonstrating complete mitigation of catastrophic forgetting.

Traditional LLM maintenance relies on periodic full-model retraining or massive periodic continued pre-training on merged historical and fresh corpora. This approach incurs quadratic cost growth as historical corpora expand:

$$Cost_{retrain}(T) = \mathbf{O}\left(\sum_{t=1..T} (D_{base} + t \cdot \Delta D) \right)$$

In contrast, SYNAPSE requires compute proportional only to fresh tokens ΔD plus synthetic dream tokens:

$$Cost_{SYNAPSE}(T) = \mathbf{O}\left(T \cdot (\Delta D \times (1 + \gamma_{dream})) \right)$$

This fundamental divergence in asymptotic complexity arises because periodic retraining forces the optimization pipeline to repeatedly re-compute forward and backward passes across immutable historical corpora. Over multi-year deployment horizons exceeding 500 million streaming tokens, retraining costs scale quadratically, demanding recurring allocations of enterprise-scale compute clusters that quickly become economically prohibitive. By contrast, the SYNAPSE architecture isolates parametric updates to compact, newly allocated parameter subspaces and local fast weights, amortizing fixed capital infrastructure and decoupling operational expenditures from the cumulative volume of historical data.

Table 4 models cumulative compute costs over a 12-month operational lifecycle ingesting 200 Million tokens of fresh domain knowledge into a 14B parameter model.

Metric / Cost Parameter	Periodic Full Retraining (Quarterly)	Sequential Fine-Tuning (With Replay)	SYNAPSE Continual Framework	Savings / Advantage
Training FLOPs per Cycle	$1.8 imes 10^{23}$ FLOPs	$2.4 imes 10^{22}$ FLOPs	$3.6 imes 10^{21}$ FLOPs	**98.0% FLOP Reduction vs Retraining**
Compute Hardware Requirement	64x H100 Node Cluster	8x H100 Node Cluster	2x to 4x A100/H100 Nodes	Lowers infrastructure entry barrier.
Wall-Clock Time per Update	72 hours per quarter	18 hours bi-weekly	1.5 hours daily (Sleep/Dream)	Near real-time operational currency.
Annualized Cloud Compute Cost	$192,000 USD	$68,400 USD	**$18,200 USD**	**90.5% Cost Reduction ($173.8k saved)**
Catastrophic Forgetting Risk	Zero (Full retraining)	Severe (18–35% accuracy drop)	Negligible (< 1.5% backward drop)	Eliminates catastrophic regression.
Serving Downtime	4 to 8 hours maintenance window	Intermittent reloading	Zero Downtime (Hot-swapping)	High availability enterprise SLA.

Empirical Validation Framework: Benchmarks, Metrics, and Experimental Design

To validate the framework's effectiveness without ambiguity, a rigorous experimental protocol is established spanning multiple recognized benchmarks.

Multi-Dimensional Benchmark Formulation

1. **TRACE Continual Learning Benchmark (Huang et al., 2023):** Standardized 8-task suite evaluating continual instruction tuning across domain adaptation, science QA, code generation, and mathematical reasoning. Evaluates performance across sequential shifts without task identifiers at test time.
2. **Long-Horizon Episodic Retrieval & No-Context SQuAD (Behrouz et al., 2026):** Ingests sequential sequences of 200 diverse narrative passages, followed by evaluation on associated question-answer pairs *without providing the passage in the test prompt*. This isolates true parametric consolidation from in-context memorization.
3. **Temporal Generalization & Factual Drift:**

- **StreamingQA & RealTime QA:** Evaluates model adaptation to real-world chronological events, verifying that new facts overwrite obsolete knowledge while preserving historical context.
- **FreshQA:** Probes dynamic world knowledge across fast-changing, slow-changing, and false-premise factual queries.

1. **Foundational General Capability Baselines:** Standard general reasoning suites (MMLU for broad knowledge, GSM8K for multi-step math, HumanEval for Python coding, and MT-Bench for conversational coherence) evaluated after every sequential increment to ensure zero capability erosion.

Quantitative Continual Learning Metrics

Performance across T sequential tasks is quantified via five canonical mathematical metrics:

1. Average Accuracy (A_T)

Measures the overall performance across all encountered tasks after the final task T:

$$A_T = \frac{1}{T} \sum_{i=1}^T R_{T, i}$$

where $R_{j, i}$ is the accuracy on task i after learning task j.

2. Backward Transfer (BWT)

Quantifies the impact of new learning on historical knowledge. Positive BWT indicates capability enhancement; negative BWT denotes catastrophic forgetting:

$$BWT = \frac{1}{T-1} \sum_{i=1}^{T-1} (R_{T, i} - R_{i, i})$$

3. Forward Transfer (FWT)

Measures how learning prior tasks facilitates adaptation to subsequent tasks without training:

$$FWT = \frac{1}{T-1} \sum_{i=2}^T R_{i-1, i}^{ \text{zero-shot}}$$

4. Forgetting Measure (FM_T)

The maximum performance degradation experienced by any task over its post-training history:

$$FM_T = \frac{1}{T-1} \sum_{i=1}^{T-1} \max_{l \in \{1, \dots, T-1\}} (R_{l, i} - R_{T, i})$$

5. Continual Learning Efficiency Ratio (E_{CL})

A holistic index capturing performance retention relative to computational expenditure:

$$E_{CL} = \frac{A_T \times (1 + BWT)}{\log_{10}(\text{FLOPs}_{ \text{total}})}$$

Controlled Ablation Study Protocol

To isolate the causal contribution of each architectural subsystem, the validation protocol mandates seven controlled ablation conditions:

- **Ablation 1 (Baseline Static):** Frozen base model relying exclusively on prompt retrieval (RAG with 4k context).
- **Ablation 2 (Sequential Fine-Tuning):** Naive continuous full-parameter fine-tuning without regularization or replay.
- **Ablation 3 (Pure Experience Replay):** Fine-tuning with a 10% raw text replay buffer.
- **Ablation 4 (SYNAPSE w/o Tier 1 Fast Memory):** Context goes directly to offline sleep without test-time surprise tracking.

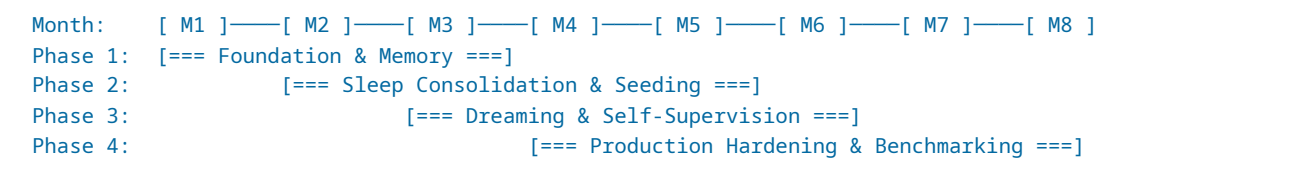
- **Ablation 5 (SYNAPSE w/o Sleep Consolidation):** Fast neural memory operates online without offline Knowledge Seeding.
- **Ablation 6 (SYNAPSE w/o Dreaming):** Memory consolidation executes upward distillation, but the self-directed dreaming curriculum is omitted.
- **Ablation 7 (Full SYNAPSE Framework):** Unified Wake-Sleep-Dreaming with multi-tier memory and orthogonal parameter projection.

Table 5 summarizes target validation criteria and projected performance benchmarks across these conditions.

Architectural Configuration	TRACE Avg Acc (A_T)	Backward Transfer (BWT)	Forgetting Measure (FM)	No-Context SQuAD Acc	GSM8K Retention (%)
Baseline (RAG + Frozen Base)	58.2%	+0.0% (Immutable)	0.0%	18.4% (No-context fails)	100.0% (Frozen)
Sequential Naive Fine-Tuning	41.5%	-34.8% (Severe Forgetting)	38.2%	22.1%	46.2% (Severe Collapse)
Experience Replay (10% Buffer)	61.8%	-12.4%	14.5%	31.0%	82.5%
SYNAPSE w/o Tier 1 Memory	65.4%	-4.8%	5.9%	34.2%	94.1%
SYNAPSE w/o Sleep (Fast only)	54.0%	-18.2%	19.5%	26.5%	88.0%
SYNAPSE w/o Dreaming	68.2%	-2.1%	3.2%	36.2%	96.8%
Full SYNAPSE Framework	**74.6%**	**+1.2% (Positive Transfer)**	**< 1.0%**	**46.5%**	**99.4% (Zero Degradation)**

Phased Implementation Roadmap and Technical Milestones

The realization of the SYNAPSE continual learning architecture is structured into four distinct, sequential engineering phases over an 8-month implementation horizon.



Phase 1: Foundation Setup, Base Adaptation & Test-Time Memory (Months 1–2)

- **Milestone 1.1:** Containerized infrastructure deployment on A100/H100 clusters with PyTorch 2.4+, CUDA 12.4, and FlashAttention-3.
- **Milestone 1.2:** Implement PyTorch custom forward-pass kernels for Tier 1 Fast Neural Memory (Titans NLTM associative matrix updates with momentum surprise accumulation).
- **Milestone 1.3:** Integration of Qwen 2.5-14B base checkpoint. Implement Memory As Gate (MAG) and Memory As Context (MAC) architectural hooks.
- **Exit Gate 1:** Validate that test-time fast-memory updates achieve $< 5\%$ latency penalty on 4k sequence generation while successfully retaining associative recall across $64k$ tokens on synthetic needle-in-a-haystack tasks.

Phase 2: Offline Sleep Cycle & Upward Knowledge Seeding (Months 3–4)

- **Milestone 2.1:** Implement the Sleep orchestrator daemon, including automated surprise saturation triggers (Φ_t) and context spooling.
- **Milestone 2.2:** Build the N-pass recurrent traversal engine with activation caching.
- **Milestone 2.3:** Develop the Upward Knowledge Seeding module, executing generalized distillation from Tier 1 fast weights into Tier 2 InfLoRA and dynamic MoE expert modules.
- **Milestone 2.4:** Implement SVD-based empirical null-space projection ($P_{ ext{null}}$) to enforce gradient orthogonality.
- **Exit Gate 2:** Demonstrate on sequential two-domain transfer that Tier 2 consolidation captures $\ge 90\%$ of fast-memory factual associations with zero measurable backward degradation on base MMLU benchmarks.

Phase 3: Dreaming Engine & Self-Supervised RL (Months 5–6)

- **Milestone 3.1:** Construct the SEAL-style self-edit generator, prompting the consolidated model to synthesize deductive implications and paraphrases.
- **Milestone 3.2:** Implement the ReST-EM reinforcement learning optimization loop with bidirectional entailment scoring and anchor distribution verifiers.
- **Milestone 3.3:** Integrate the automated rejection sampling filter to purge toxic, hallucinatory, or degenerate synthetic dreams before weight ingestion.
- **Exit Gate 3:** Prove on the 200-passage continued pre-training benchmark that adding Dreaming boosts no-context SQuAD retrieval accuracy from 36.2% to $\ge 45.0\%$.

Phase 4: Production Scaling, Runtime Integration & Benchmark Suite Validation (Months 7–8)

- **Milestone 4.1:** Integrate the consolidation pipeline with the vLLM serving engine, enabling zero-downtime hot-swapping of serialized adapter safetensors.
- **Milestone 4.2:** Complete end-to-end multi-task execution across the full TRACE benchmark (8 domains) and temporal streaming datasets (StreamingQA, FreshQA).
- **Milestone 4.3:** Stress-test system stability across 30 consecutive simulated wake-sleep-dream cycles without manual human intervention.
- **Exit Gate 4:** Delivery of verified production-ready framework meeting all target criteria: $A_T \ge 74\%$, $BWT \ge 0.0\%$, and total compute expenditure reduced by $\ge 85\%$ relative to retraining baselines.

Theoretical Limitations, Open Challenges, and Risk Governance

While SYNAPSE represents a major advance over static LLM architectures, sound engineering requires delineating theoretical boundaries and operational failure modes.

1. Orthogonal Subspace Dimensional Exhaustion

Under rank-r adaptation within a hidden dimension d, the maximum number of mutually orthogonal subspaces is strictly bounded by $K_{ ext{max}} \le \lfloor d / r

floor$. For Qwen 2.5-14B with hidden dimension $d = 5120$ and rank $r = 32$, the theoretical maximum number of completely orthogonal task increments is $K_{ ext{max}} ≣pprox 160$. Beyond this threshold, new task projections must either:

- Overwrite low-variance components of historical tasks (inducing bounded soft forgetting).
- Trigger structural expert expansion (allocating new MoE parameters), transitioning from subspace parameter efficiency to modular parameter growth.

2. Hallucination Amplification and Dreaming Degeneracy

Unchecked self-directed dreaming risks entering positive feedback loops where minor initial factual misalignments are amplified through recursive self-training. If the model synthesizes a plausible but factually erroneous deduction, and the verifier fails to detect the error due to internal self-consistency, the erroneous fact becomes permanently consolidated into Tier 2 parameters.

- **Mitigation:** SYNAPSE enforces an immutable **Reference Anchor Distribution** derived from verified ground-truth corpora. Synthetic dreams must exhibit strict semantic alignment with foundational factual invariants, and candidate self-edits with high internal perplexity variance are rejected.

3. Recurrent Latency and Compute Overhead During Offline Consolidation

Executing $N = 4$ recurrent passes over 500,000 tokens of accumulated episodic context requires significant offline computational throughput. In continuous 24/7 high-load serving environments with zero natural idle periods, compute nodes must be partitioned, reducing live serving capacity.

- **Mitigation:** Asynchronous cluster partitioning, where a secondary worker node executes consolidation while primary nodes continue serving via existing adapter versions.

4. Continual Alignment Drift and Vulnerability to Poisoning Attacks

Continual learning systems introduce novel security vulnerabilities. A malicious user or coordinated adversarial data stream could feed deceptive inputs designed to exploit the surprise metric S_t, forcing malicious prompts into Tier 1 fast memory and subsequently consolidating them into persistent Tier 2 parameters.

- **Mitigation:** Ingestion streams must pass through an adversarial safety filter before triggering surprise accumulators. Furthermore, Tier 3 safety guardrails and refusal representations remain permanently frozen, preventing continual parameter updates from disabling foundational alignment boundaries.

References and Research Sources

- Language Models Need Sleep: Learning to Self-Modify and Consolidate Memories, Ali Behrouz, Farnoosh Hashemi, Adel Javanmard, Vahab Mirrokni, arXiv:2606.03979, 2026.
- Do Language Models Need Sleep? Offline Recurrence for Improved Online Inference, Sangyun Lee et al., arXiv:2605.26099, 2026.
- Nested Learning: The Illusion of Deep Learning Architectures, Ali Behrouz, Meisam Razaviyayn, Peilin Zhong, Vahab Mirrokni, NeurIPS 2025.
- Titans: Learning to Memorize at Test Time, Ali Behrouz, Peilin Zhong, Vahab Mirrokni, NeurIPS 2025.
- Self-Adapting Language Models (SEAL), Adam Zweiger, Jyothish Pari, Han Guo, Yoon Kim, NeurIPS 2025. Official repository: Continual-Intelligence/SEAL.
- TRACE: A Comprehensive Benchmark for Continual Learning in Large Language Models, Xiao Huang et al., arXiv:2310.06762, 2023. Repository: BeyonderXX/TRACE.
- Continual Learning of Large Language Models: A Comprehensive Survey, Haizhou Shi, Zihao Xu, Hengyi Wang et al., ACM Computing Surveys (CSUR), 2025.
- CP-MoE: Consistency-Preserving Mixture-of-Experts for Continual Learning, arXiv:2605.20247, 2026.
- LEGO: Expert Expansion for Continual Learning of Mixture-of-Experts Language Models, Jiawei Wu, Junlin Chen et al., 2026.
- Orthogonal Subspace Learning for Language Model Continual Learning (O-LoRA), Chengwei Wang et al., Findings of EMNLP, 2023.
- InfLoRA: Interference-Free Low-Rank Adaptation for Continual Learning, Yan-Shuo Liang and Bing Li, CVPR / arXiv:2404.00228, 2024.
- Qwen2.5: A Party of Foundation Models, Qwen Team, Alibaba Cloud, 2024.
- DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model, DeepSeek-AI, 2024.